



SQL

INTERVIEW QUESTION

by Pycode.Hubb



Q1. What is SQL?

- SQL (Structured Query Language) is used to store, retrieve, update, and manage data in databases.
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Q2. What is DBMS?

- DBMS (Database Management System) is software used to create, manage, and control databases.
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Q3. What is RDBMS?

- RDBMS (Relational Database Management System) stores data in tables and supports relationships between tables.
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Q4. Difference Between DBMS and RDBMS?

- DBMS manages data without table relationships, whereas RDBMS manages related data using keys.
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Q5. What is a Table?

- A table is a collection of rows and columns used to store data.
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Q6. What is a Primary Key?

- A primary key uniquely identifies each record in a table and cannot contain NULL values.
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Q7. What is a Foreign Key?

➔ A foreign key links one table to another by referencing a primary key.

Q8. What is a Unique Key?

➔ A unique key ensures that all values in a column are distinct.

Q9. What is a NOT NULL Constraint?

➔ It ensures that a column cannot contain NULL values.

Q10. What is a DEFAULT Constraint?

➔ It assigns a default value when no value is provided.

Q11. SQL Query to Fetch All Data

➔ `SELECT * FROM employees;`

Q12. Difference between DELETE, TRUNCATE, and DROP?

➔ DELETE removes selected rows, TRUNCATE removes all rows, and DROP removes the entire table.



Q13. What is the WHERE Clause?

- ➔ The WHERE clause filters records based on a specified condition.

```
SELECT * FROM employees WHERE salary > 50000;
```

Q14. What is the HAVING Clause?

- ➔ The HAVING clause filters grouped data after aggregation.
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Q15. What is GROUP BY?

- ➔ GROUP BY groups rows with the same values into summary results.
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Q16. What is ORDER BY?

- ➔ ORDER BY sorts query results in ascending or descending order.
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Q17. What is JOIN?

- ➔ JOIN combines data from multiple tables based on a related column.
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Q18. What are the Types of JOIN?

- ➔ The main types are INNER JOIN, LEFT JOIN, RIGHT JOIN, FULL JOIN, CROSS JOIN, and SELF JOIN.
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Q19. What is INNER JOIN?

➔ INNER JOIN returns only matching records from both tables.

Q20. What is LEFT JOIN?

➔ LEFT JOIN returns all records from the left table and matching records from the right table.

Q21. What is RIGHT JOIN?

➔ RIGHT JOIN returns all records from the right table and matching records from the left table.

Q22. What is FULL JOIN?

➔ FULL JOIN returns all matching and non-matching records from both tables.

Q23. What is SELF JOIN?

➔ SELF JOIN is a table joined with itself.

Q24. What is CROSS JOIN?

➔ SELF JOIN is a table joined with itself. CROSS JOIN returns every possible combination of rows from both tables.



Q25. What is UNION?

- ➔ UNION combines results from multiple queries and removes duplicate rows.
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Q26. Difference between UNION and UNION ALL?

- ➔ UNION removes duplicates, while UNION ALL keeps duplicates.
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Q27. What is a Subquery?

- ➔ A subquery is a query written inside another query.
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Q28. What is a Correlated Subquery?

- ➔ A correlated subquery depends on the outer query and executes once for each row.
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Q29. What is Normalization?

- ➔ Normalization organizes data to reduce redundancy and improve consistency.
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Q30. What are the Types of Normalization?

- ➔ The common types are 1NF, 2NF, 3NF, and BCNF.
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Q31. What is Denormalization?

- ➔ Denormalization adds redundancy to improve query performance.
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Q32. What is an Index?

- ➔ An index improves the speed of data retrieval operations.
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Q33. What is the Drawback of an Index?

- ➔ Indexes improve read performance but can slow down insert, update, and delete operations.
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Q34. What is a View?

- ➔ A view is a virtual table created from the result of a query.
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Q35. What is a Stored Procedure?

- ➔ A stored procedure is a precompiled set of SQL statements stored in the database.
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Q36. What is a Trigger?

- ➔ A trigger automatically executes when a specified database event occurs.
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Q37. What is a Cursor?

➔ A cursor processes query results one row at a time.

Q38. What is a Transaction?

➔ A transaction is a group of SQL operations executed as a single unit of work.

Q39. What are ACID Properties?

➔ ACID stands for Atomicity, Consistency, Isolation, and Durability.

Q40. What is COMMIT?

➔ COMMIT permanently saves changes made during a transaction.

Q41. What is ROLLBACK?

➔ ROLLBACK reverses changes made during a transaction.



Q42. What is SAVEPOINT?

➔ A SAVEPOINT allows partial rollback within a transaction.

Q43. What is SQL Injection?

➔ SQL Injection is a security attack that inserts malicious SQL code into user input.

Q44. How to Prevent SQL Injection?

➔ Use prepared statements, parameterized queries, and input validation.

Q45. Difference between CHAR and VARCHAR?

➔ CHAR stores fixed-length data, while VARCHAR stores variable-length data.

Q46. What is NULL?

➔ NULL represents a missing, unknown, or unavailable value.



Q47. Difference between COUNT(*) and COUNT(column)?

➔ COUNT(*) counts all rows, while COUNT(column) counts only non-NULL values.

Q48. What is a Primary Key?

➔ A primary key uniquely identifies each row in a table.

Q49. What is a Foreign Key?

➔ A foreign key references a primary key in another table to maintain relationships.

Q50. What is Referential Integrity?

➔ Referential integrity ensures relationships between tables remain consistent.

Q51. What is a Constraint?

➔ A constraint is a rule that enforces data accuracy and integrity.



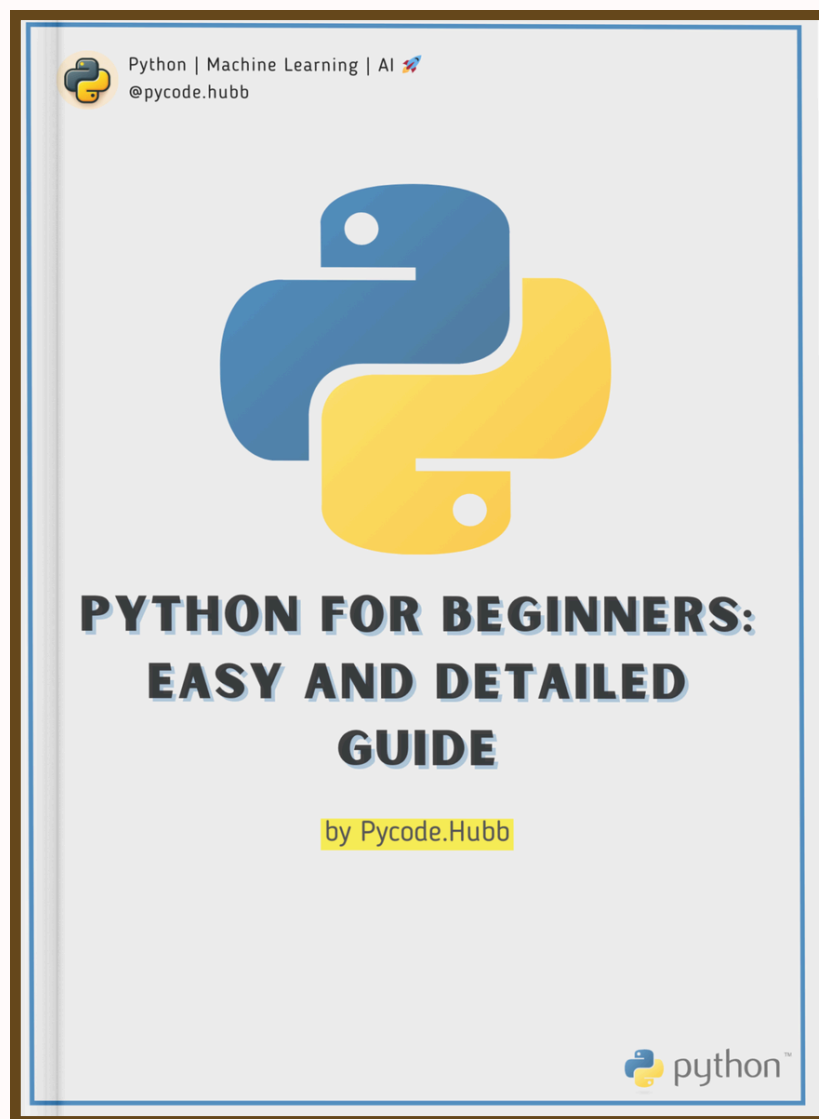
Q52. Difference between DELETE, TRUNCATE, and DROP?

➔ DELETE removes selected rows, TRUNCATE removes all rows while keeping the table, and DROP removes the entire table structure and data.



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